Tutorial 4

Research design exercises

Task 1. Form behavioural research questions in usability testing context:

Behavioural research questions on the usability of a product focus on how users interact with a product, the problems they encounter, and the overall user experience. These questions help identify usability issues and areas for improvement. Here is a list of issues around which you can form behavioural research questions on product usability:

- Navigation and Findability
- Error Identification and Recovery
- Task Completion
- Learning Curve
- Mobile Usability
- Form and Data Entry
- Feedback and Help-Seeking Behaviour
- Content Interaction
- Task Priority and Sequencing
- Search Behaviour
- Multimodal Interaction
- Task Switching and Multitasking
- User Preferences
- User Satisfaction and Emotional Responses

Work in pairs. Pick one of the topics above and create a research question.

Make sure that the research question is **clear**, **feasible**, **specific** and **measurable**. This can be achieved if the research question is written so that it outlines various aspects of the study, including:

- ❑ the **population**
- ❑ the **variables** to be studied and
- ❑ the **problem the study addresses**

Type in here your research question
"What effect does screen size have on task completion?"

Share your question to be discussed with the class.

Based on the discussion and the comments you received revise your research question.

Rewrite your research question
"What effect does the screen size of mobile devices have on the task completion time for e-commerce transactions among users aged 50 and above?"

Task 2. Review the following hypotheses:

Read the following hypotheses and complete the following tasks:

- identify the **dependent and independent variables** in each hypothesis
- identify the **treatments, or conditions** refer to the different techniques, devices, or procedures that we want to compare
- identify the **units** (the objects to which we apply the experiment treatments, in UX context the units are normally human subjects with specific characteristics, such as gender, age, or computing experience)
- identify the **assignment method,** the way in which the experimental units are assigned different treatments

H1. There is no difference in users' reading speed and retention rate when they view news on a desktop computer or a mobile.

H2. There is no difference in the mean selection time across joystick, touch, and gesture.

H3. There is no difference in the technology adoption rate between two speech-based applications with different dialog designs.

H4. There is no difference in the reading skills of children who used educational software for 6 months compared to those who have never used the software.

H1. There is no difference in users' reading speed and retention rate when they view news on a desktop computer or a PDA.

Dependant variables: … [reading speed & retention rate]

Independent variables: … [the device used for the task completion]

Treatment/conditions: … [We have two conditions in the experiment, one testing the reading speed on desktop devices and one on mobile]

Units: …. [users to be included in the study, here we need to specify the exact characteristics of the users to be recruited for the study]

Assignment method: …. [between group, one group will read a piece of text on a desktop devices and another group will read the same piece of text the on a PDA, the style of the text will be the same in both conditions]

H2. There is no difference in the target selection speed and error rate between joystick, touch screen, and gesture recognition.

Dependant variables: … [target selection speed & error]

Independent variables: … [the device used for the task completion joystick, touch screen, and gesture recognition]

Treatment/conditions: … [users completing a task using joystick, touch screen, and gesture recognition]

[We have three conditions in the experiment, completing the task using joystick, touch screen, and gesture recognition]

Units: …. [users to be included in the study, here we need to specify the exact characteristics of the users to be recruited for the study]

Assignment method: …. [between group, one group will complete a task using a joystick, a touchscreen and gesture recognition

We could use within group assigning asking each user to complete three different tasks of equal level of difficulty using the three different devices]

H3. There is no difference in the technology adoption rate between two speech based applications with different dialog designs.

Dependant variables: … [technology adoption rate]

Independent variables: … [two speech-based applications with different dialog designs]

Treatment/conditions: … [users completing a task following two speech-based dialog design A and B]

We have two conditions in the experiment, users completing a task following speech-based dialog design A and B]

Units: …. [users to be included in the study, here we need to specify the exact characteristics of the users to be recruited for the study]

Assignment method: …. [between group, one group will complete a task speech-based dialog design A and the other using design B

We could use within group assigning asking each user to complete different tasks of equal level of difficulty using speech-based dialog design A and B

It would require randomization in the order to avoid familiarity of the users with the task]

H4. There is no difference in the reading skills of children who used educational software for 6 months compared to those who have never used the software.

Dependant variables: … [reading ability]

Independent variables: … [the use of educational software for 6 months, no use of software].

Treatment/conditions: … [children developing reading skills using an educational software for 6 months, children developing reading skills without using an educational software for 6 months]

We have two conditions children developing reading skills with and without using educational software for 6 months]

Units: …. [users to be included in the study, here we need to specify the exact characteristics of the users to be recruited for the study]

Assignment method: …. [between group, one group of children will developing reading skills using an educational software for 6 months, another group of children with the same characteristics will develop reading skills without using an educational software for 6 months, then both groups will be exposed to a test condition to review their reading skills]

Task 3. Form a research hypothesis.

Form a hypothesis that will allow you to study the usability of a product. Focus on how users interact with a product, the problems they encounter, and the overall user experience.
Form both the Null and the Alternative hypothesis.

Remember a **hypothesis is a declarative statement about the relationship between two or more variables based on statistical consideration.**

Here is one example:

*RQ: "Among banking customers performing the same transfer task, does using the **mobile app** versus the **web platform** lead to **different task-completion times**?"*

In this hypothesis, you are proposing that there will be a significant difference in task completion times between users of the mobile app and the web-based platform when performing a specific task (transferring funds). This hypothesis sets the stage for a study that will investigate and measure the behavioural differences in usability between the two versions of the product.

Null Hypothesis (H_0): *"Users of our mobile banking app will not exhibit significantly faster task completion times when transferring funds between accounts compared to users of our web-based banking platform."*
or
"Users take the **same average time** to transfer funds on the mobile app and on the web platform."

Alternative Hypothesis (H_1): *"Users of our mobile banking app will exhibit significantly faster task completion times when transferring funds between accounts compared to users of our web-based banking platform."*
or
*"Users take a **different average time** to transfer funds on the mobile app versus the web platform."*

The mathematical expression of this hypothesis is as follows:

- H_0: $\mu_m = \mu_w$, or H_0: $\mu_m - \mu_w = 0$, or H_0: $\mu_m - \mu_w = 0$
- H_1: $\mu_m \neq \mu_w$

μ_m = the mean of mobile banking task completion

μ_w = the mean of web banking task completion

Work in pairs. Pick one of the topics in task 1 and create a research hypothesis or use the research question you have formed in task 1 and turn this in research hypothesis.

Make sure that the hypothesis is written so that it outlines various aspects of the study states:

- the **dependent and independent variables**
- the **treatments, or conditions**
- the **units**
- and it implies the **assignment method**

Type in here the Null and the Alternative research hypothesis
Null Hypothesis (H0): *"Users of our mobile banking app will not exhibit significantly faster task completion times when transferring funds between accounts compared to users of our web-based banking platform."* **Alternative Hypothesis (H1):** *"Users of our mobile banking app will exhibit significantly faster task completion times when transferring funds between accounts compared to users of our web-based banking platform."*

Share your question to be discussed with *the* class.

Based on the discussion and the comments you received revise your research question.

Rewrite your research hypothesis .....

Try something else, use **Copilot** as a tool to review the correctness of your hypothesis.

Rewrite your research hypothesis based on **Copilot's** recommendation. Compare this with the reviews you received from your peers. .....

In your refined hypothesis clearly specify the following:

- the **dependent and independent variables** of the hypothesis
- the **treatments, or conditions**
- the **units**
- the suggested **assignment method you will use and why**

Task 4. Experiment design

Read the following scenarios. Discuss the type of experiment design (between-group, within-group, or split-plot) that is appropriate for each scenario.

Scenario 1: A study investigates whether people who have attended a security training program generate and use more secure passwords than people who have not received any security training.

Scenario 2: A research team examines the effectiveness of joysticks and trackballs for selecting static targets and moving targets.

Scenario 3: A research team examines whether the gender of an online agent affects the perception of trust for young users, middle-aged users, and older users.

Scenario 4: A research team examines whether virtual teams who use video conferencing are more productive than teams who use phone-based teleconferencing.

Scenario 5: A study examines the effectiveness of three menu structures. The first structure has two levels, with 8 items in the first level and 64 items in the second level. The second structure has three levels, with 4 items in the first level, 16 items in the second level, and 64 items in the third level. The third menu has six levels, with 2 items in the first level and 2 n items in the n the level.

Provide your answer for each scenario:

Scenario 1: A study investigates whether people who have attended a security training program generate and use more secure passwords than people who have not received any security training.

Your answer: Between- group

In this type of design, participants are divided into two separate groups: one group that has received the security training program, and another group that has not received any security training. The groups are then compared to determine if there are differences in the generation and use of secure passwords between the two groups.

This design is appropriate when you want to assess the impact of an independent variable (in this case, attendance of a security training program) on a dependent variable (the generation and use of secure passwords) by comparing different groups of participants who have been exposed to different conditions.

Scenario 2: A research team examines the effectiveness of joysticks and trackballs for selecting static targets and moving targets.

Your answer: Within- subjects design

For the scenario described, where you are examining the effectiveness of two different input devices (joysticks and trackballs) for selecting both static and moving targets, you would typically use a within-group study design. This is also known as a repeated measures design or a **within-subjects design**.

In a within-group design, the same group of participants is exposed to both conditions (in this case, using joysticks and trackballs) to evaluate the impact of the independent variable (input device) on the dependent variable (target selection for both static and moving targets).

This design allows you to directly compare the performance of the same individuals under different conditions and is particularly useful when you want to assess how participants' performance varies when exposed to various treatments or conditions.

To avoid users having become familiar with the task you have to use randomization of the subjects.

Scenario 3: *A research team examines whether the gender of an online agent affects the perception of trust for young users, middle-aged users, and older users.*

Your answer: Split-plot design

A split-plot design is a hybrid experimental design that combines both within-group (repeated measures) and between-group (independent measures) factors. In your scenario:

The age groups (young users, middle-aged users, and older users) are treated as a **between-group** factor because you are interested in comparing how different age groups respond to the gender of the online agent. Each participant belongs to one of these age groups.

The **gender of the online agent** (e.g., male agent and female agent) is treated as a **within-group factor** because participants (groups of users belonging to different age groups) will be exposed into different conditions (e.g., male agent and female agent)

In a **split-plot design**, participants from each age group (within-group factor) are exposed to both levels of the **between-group factor** (gender of the online agent) to assess the impact of each factor and their interaction. This allows you to analyze how trust perception varies based on both age and gender conditions.

A **split-plot design** is well-suited when you want to investigate the influence of both within-group and between-group factors on your dependent variable, which is the perception of trust in this case.

Scenario 4: A research team examines whether virtual teams who use video conferencing are more productive than teams who use phone-based teleconferencing.

Your answer: Between-group design

For the described scenario, where you want to compare the productivity of virtual teams using two different communication methods (video conferencing and phone-based teleconferencing), you would use a **between-group study design**.

In a **between-group design**, you create separate groups for each condition you are studying. In this case, you would have one group using video conferencing and another group using phone-based teleconferencing. Each group represents a different condition (communication method), and you compare their productivity to determine if there are differences.

This design is suitable when you want to assess the impact of an independent variable (communication method) on the dependent variable (productivity) by comparing different groups of participants exposed to different conditions. It allows you to determine whether one communication method is more productive than the other.

Scenario 5: A study examines the effectiveness of three menu structures. The first structure has two levels, with 8 items in the first level and 64 items in the second level. The second structure has three levels, with 4 items in the first level, 16 items in the second level, and 64 items in the third level. The third menu has six levels, with 2 items in the first level and 2 n items in the n the level.

Your answer: Within- subjects design

For the described scenario, where you are examining the effectiveness of three different menu structures with varying levels and items, you would typically use a within-group study design. This is also known as a repeated measures design or a **within-subjects design**.

In a within-group design, the same group of participants is exposed to multiple conditions. In your case, participants would interact with all three menu structures. This design allows you to directly compare how the same individuals respond to different treatments or conditions.

In your specific scenario, participants would experience three different menu structures (the independent variable), and you would measure and compare their performance, preferences, or effectiveness across these structures (the dependent variable).

This within-group design is appropriate when you want to assess how participants' behavior and preferences change when exposed to various conditions or treatments, which is the case when comparing different menu structures.